

- Filename: comptia-pentest-plus-pt0-002-3-2-1-denial-of-service.md
- Show Name: Pentest+ (PT0-002)
- Topic Name: Attacks and Exploits
- Episode Name: Denial of Service

Denial of Service

Objectives:

- Define Denial of Service and Distributed Denial of Service attacks
- Define Reflection and Amplification DoS/DDoS attacks
- Explain and demonstrate Volumetric, Protocol, and Application

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- DoS and DDoS
 - Reflection and Amplification
 - Reflection
 - Attacker spoofs Target IP and requests information
 - "Trigger Packet"
 - Requested info is then sent to Target
 - Use Botnet for DDoS
 - Amplification
 - Like a reflection attack, but...
 - Requested info payload is much larger than the Trigger Packet
 - Examples include: *DNS* and *NTP*
 - Volumetric vs Protocol vs Application
 - Volumetric
 - ICMP Flood
 - Reflection networks are useful here
 - Requests and Replies saturates the network
 - `hping3 --flood --spoof 192.168.241.133 --icmp -p 53 192.168.241.130`
 - Smurf & Fraggle
 - DDoS target with ICMP echo replies
 - Send ICMP echo request to network broadcast address with spoofed source IP of target
 - If network allows directed broadcast requests, all hosts on network will respond to target with ICMP echo replies
 - `hping3 --flood --spoof --icmp 192.168.241.130 192.168.241.255`
 - Fraggle
 - Similar to Smurf
 - UDP instead of ICMP
 - Targets ports 7(Echo) and 19(CHARGEN)

- Protocol

- Ping of Death

- Oversized packet
 - Size greater than 65535 bytes
 - System crashes

- UDP Flood

- Attacker floods target with UDP packets from spoofed IP
 - Target checks for UDP service
 - Target responds with 'Destination Unreachable' error
 - Legit traffic can't access server
 - `hping3 --flood --spoof 192.168.241.133 --udp -p 53 192.168.241.130`

- Application

- HTTP

- Web services utilize HTTP Methods
 - GET
 - POST
 - Attacker overwhelms Server by flooding it with GET/POST requests

- **LOIC DEMO**

- Unintentional attack

- Death of Michael Jackson
 - Stopped, slowed and even crashed
 - Google
 - Twitter
 - News/Link sites
 - Link goes viral and site gets overloaded by visits
 - Reddit Hug of Death
 - Digg Effect

- SlowLoris

- **METASPLOIT DEMO**

- Read description in 'info' section